

Assignment : A+- Troubleshooting and helpdesk

Day 5: Case Study Practical (1 hour): Work on a scenario involving multiple hardware issues.

TASK : 1

A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.



Problem:

The user reports that the laptop does not power on even when the charger is connected.

Troubleshooting Steps:

1. Check the power socket:

First, check whether the wall socket is working by connecting another working device to it. This helps determine whether the problem is with the power source.

2. Check the charger:

Inspect the charger adapter, cable, and connector for cuts, damage, loose connections, or signs of overheating. A faulty charger may not supply power to the laptop.

3. Check the charging port:

Inspect the laptop's charging port for dust, physical damage, or a loose connection. Make sure the charger connector fits properly.

4. Check the battery:

If the laptop has a removable battery, inspect it for visible damage and check whether it is properly connected. A faulty or completely discharged battery can prevent the laptop from starting.

5. Perform a power reset:

Disconnect the charger and, if possible, the battery. Hold the power button for about 15–20 seconds. Reconnect the appropriate power source and try turning on the laptop again. This can clear a temporary power-state problem.

Conclusion:

A laptop that does not power on can have problems with the power socket, charger, charging port, battery, or internal power circuitry. Troubleshooting should begin with simple external checks before moving to internal hardware diagnosis.

Task : 2

You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.



Problem:

The user reports that the monitor is showing a blank screen, but the CPU appears to be running.

Troubleshooting Flow:

1. Check monitor power:

Make sure the monitor is switched on and its power cable is connected properly. Check whether the power indicator is working.

2. Check the display cable:

Check the HDMI, DisplayPort, or VGA cable at both ends. Make sure it is firmly connected and not damaged. If possible, test with another working cable.

3. Check the input source:

Make sure the monitor is set to the correct input source, such as HDMI or DisplayPort, according to the cable being used.

4. Check the GPU connection:

If the computer has a dedicated graphics card, make sure the monitor cable is connected to the GPU's display output. Also check that the GPU is properly seated and its power connectors are connected.

5. Check POST and RAM:

If there is still no display, check for POST beep codes or motherboard diagnostic indicators. A RAM problem can also result in a computer starting without displaying anything.

6. Check display settings:

If the display works during startup but becomes blank after the operating system loads, check the display resolution, refresh rate, display mode, and selected monitor.

7. Confirm the solution:

Restart the computer and confirm that the monitor displays the startup screen and operating system normally.

Conclusion:

A blank monitor can be caused by a power problem, faulty or loose cable, incorrect input source, GPU connection, RAM/POST problem, or incorrect display settings. Troubleshooting should begin with simple connections before checking internal hardware.

TASK : 3

A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.



Problem:

The wireless mouse is not responding, but the keyboard is working correctly.

Troubleshooting Steps:

1. Check the mouse power:

Check that the mouse is switched on. Replace or recharge the batteries if necessary and check whether the mouse shows any signs of power.

2. Check the wireless receiver:

Check the USB wireless receiver and make sure it is properly connected to the computer. Disconnect and reconnect it.

3. Test another USB port:

Connect the wireless receiver to another known-working USB port. If the mouse works, the original USB port may be faulty.

4. Test the mouse on another computer:

Connect the receiver to another computer and test the mouse. If it works on another computer, the original system may have a USB or configuration problem. If it does not work, the mouse, battery, or receiver may be faulty.

5. Check system settings:

If the mouse is detected but does not respond, check the mouse/device settings and make sure the device is enabled. Device Manager can also be checked for hardware or driver problems.

6. Confirm the fix:

Move the mouse and verify that the pointer moves correctly and the buttons respond.

Conclusion:

Testing the battery, wireless receiver, different USB ports, and another computer helps isolate whether the problem is with the mouse, USB port, or computer settings.

TASK : 4

Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste



Problem:

A gaming PC is overheating and shutting down during use. This can happen when the cooling system is not able to remove enough heat from the components.

1. Fan Problem:

The CPU, GPU, or case fan may not be working correctly. I would check whether the fans are spinning when the PC is running and look for unusual noise or a fan that is not rotating. A faulty fan should be cleaned, properly connected, or replaced if necessary.

2. Dust Buildup:

Dust can collect on fans, heatsinks, air vents, and filters. This can restrict airflow and cause the components to become hot. I would turn off and unplug the PC and inspect the inside for excessive dust. The dust should be cleaned using proper PC-cleaning methods.

3. Thermal Paste Problem:

Dried or insufficient thermal paste can reduce heat transfer between the CPU and its heatsink. I would check whether the heatsink is properly mounted. If required, the old thermal paste can be cleaned and replaced with suitable new thermal paste.

Conclusion:

Three common hardware causes of overheating are faulty fans, dust buildup, and thermal paste problems. Checking and maintaining the cooling system can help prevent overheating and unexpected shutdowns.